

Michael Tang

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EDUCATION

Carnegie Mellon University

M.S. in Electrical & Computer Engineering

Pittsburgh, PA

Expected May 2028

University of California, Santa Barbara

B.S. in Computer Engineering; Dean's Honors, College of Engineering

Santa Barbara, CA

June 2026

EXPERIENCE

UCSB Novel Electronic Device & Circuits Lab

Santa Barbara, CA

VLSI Undergraduate Researcher, Hardware Accelerator VLSI

Oct 2025 – Aug 2026

- Owned RTL-to-signoff implementation of a 2 mm² Rocket+RoCC accelerator for SAT-solving, driving floorplanning, power delivery, and timing closure in TSMC 28nm.
- Developed Python/TCL automation for Cadence Innovus power-grid generation and std-cell halo placement, eliminating power-strap spacing and macro placement violations.
- Executed DRC and LVS signoff using Calibre and debugging implementation violations across logic and SRAM standard-cell regions prior to tapeout.

UCSB ArchLab

Santa Barbara, CA

Microarchitecture Undergraduate Researcher, CPU Microarchitecture

Sep 2025 – Jun 2026

- Designed RTL for a gshare branch predictor in SystemVerilog within an open-source RV64GC superscalar out-of-order core to improve front-end performance.
- Developed a cocotb-based and UVM verification environment to validate functional correctness of the gshare predictor and debug RTL behavior against expected branch-resolution outcomes.

PROJECTS

Silicon Reverse Engineering | *Python, Yosys, KLayout, Z3 SMT, Physical Design*

Summer 2026

- Extracted a 92-flop gate-level netlist from raw GDS in KLayout, verifying RTL isomorphism via Icarus Verilog co-simulation.
- Backward-traced flop dependencies through 56-stage reconvergent fan-in, recovering a 121-state base-11 counter and 11 saturating key-press counters.
- Solved a 54-flop acceptance condition via Z3 SMT bounded model checking, recovering a satisfying 121-bit input sequence.
- Decoded the sequence as an 11×11 Star Battle solution, extracting a hidden 15-byte ASCII message via gate-level simulation.

ML-Based Fine-Grained RTL Timing Estimation | *Python, Yosys, OpenSTA, ML for EDA*

Spring 2026

- Developed an ML pipeline for register-level timing prediction using synthesis and timing data extracted from Yosys and OpenSTA.
- Engineered LambdaMART ranking models achieving 78.2% COVR in identifying the top 5% most timing-critical registers across 20 synthesized RTL designs.
- Built XGBoost regression models predicting global Worst Negative Slack (WNS) within 7.5% error while developing automated feature-extraction pipelines for large-scale timing datasets.

Optimized Chip Clock Tree in IBM 130nm | *HSpice, Digital IC Design*

Fall 2025

- Designed a clock distribution tree for a 5 mm² chip powering 100,000 flip-flops.
- Optimized clock jitter and rise-time to within 100 ps via buffer sizing, buffer spacing, and metal layer selection.
- Performed power analysis across 1 mm² power domains and validated performance using 1,000 Monte Carlo simulations (5th–95th percentiles).

TECHNICAL SKILLS

Languages: SystemVerilog, Verilog, Python, TCL, Bash, C/C++

EDA: Innovus, Virtuoso, Calibre, OpenSTA, Yosys, HSpice, LTSpice, ModelSim, Vivado, Icarus Verilog, cocotb

ASIC: Digital Design, RTL Synthesis, Place and Route, Power Analysis, Static Timing Analysis, Clock Tree Synthesis, Formal Verification, Gate-Level Simulation, DRC/LVS Signoff, UVM